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RESEARCH ARTICLE

Stretchable magnesium–air battery based on dual ions conducting hydrogel for intelligent biomedical applications

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Abstract

Flexible and bio-integrated electronics have attracted great attention due to their enormous contributions to personalized medical devices. Power sources, serving as one of the most important components, have been suffering from many problems, including deficient biocompatibility, poor stretchability, and unstable electrical outputs under deformed conditions, which limits the practical applications in flexible and bio-integrated electronics. Here, we reported a fully stretchable magnesium (Mg)–air battery based on dual-ions-conducting hydrogels (SDICH). The high-performance battery enables long-term operation with lighting 120 lighting emitting diodes (LEDs) for over 5 h. Benefiting from the advanced materials and mechanical designs, the battery exhibits stability electrical outputs under stretching, which allows to operate ordinarily under various mechanical deformations without performance decay. Furthermore, the great biocompatibility of the battery offers great opportunity for biomedical applications, which is demonstrated by a self-adaption wound dressing system. The in vitro and in vivo results prove that the self-adaption wound dressing can effectively prevent wound inflammation and promote wound healing. By exploiting thermal feedback mechanics, the system can adjust antibiotic release

Xingcan Huang, Yiming Liu, Wooyoung Park, and Zhao Zhao contributed equally to this study.

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rate and dosage spontaneously according to the real-time wound conditions. The proposed fully stretchable Mg–air battery and self-adaption wound dressing display great potential in skin-integrated electronics and personalized medicine.

KEYWORDS

bioelectronics, flexible electronics, Mg–air battery, stretchable battery, wound healing

1 | INTRODUCTION

The unique features in flexible electronics from excellent wearability, to miniaturized device size, to ultrathin and lightweight^{1–5} enable flexible electronics for a wide range of applications in the fields of disease diagnostics,^{6–8} human–machine interfaces,^{9–11} and healthcare monitoring.^{12–14} As an advanced format of flexible electronics, skin-integrated electronics with excellent stretchability have been spotlighted as a prospective research topic for the next generation of wearable technologies and personality medicine because of the ability to be deformed into arbitrary shapes along with skin deformation.¹⁵ Despite advances of skin-integrated electronics in health monitoring and disease therapy, most wearable biomedical devices are not able to respond to the variations of real-time body conditions, while intelligent wearable devices can well address these issues by integrating sensing and actuating systems.^{16–18} Generally, traditional intelligent biomedical devices always contain rigid power sources, complex circuit designs, and brittle chips to achieve real-time data process and transmission as well as intelligent feedback, which are not stretchable and thus would decrease the flexibility of the whole system. As a result, it is necessary to develop fully stretchable intelligent devices with stretchable power sources and self-adaption systems for close-loop health monitoring and therapies.

Power sources are indispensable components, which enable smart skin-integrated electronics to continuously function. The requirements of power sources in skin-integrated electronics include high power density, good discharging capacity and safe materials with minimum or none hazards. Conventional wearable electronics typically adopt lithium (Li)-ion based batteries as which exhibits champion electrical outputs with miniaturized size. However, the rigid substrates and materials used in Li-ion batteries fail to provide sufficient flexibility, which is essential for flexible electronics in various practical applications. Furthermore, the poor biocompatibility of

the materials (acid or alkali electrolytes) also raises concerns of safety issue to skin.^{19,20} To find a suitable power management solution, researchers have put extensive efforts on developing new power sources. A variety of flexible including self-powered technologies (i.e., triboelectric nanogenerators, TENGs and piezoelectric nanogenerators, PENGs), wireless powering technologies based on radio frequency (RF), and soft biocompatible batteries have been reported.^{21–23} These powering technologies somehow have unsolved shortcomings for operating flexible electronics. For instance, the two typical self-powered technologies, TENGs and PENGs are both limited by the low energy conversion efficiency between mechanics and electricity.^{24,25} Although flexible electronics could be further miniaturized by applying RF technologies instead of conventional bulky batteries, in the aspect of daily usage, more critical flaws are resulted due to its extremely short operation distance (<10 cm). Recent developments in soft biocompatible batteries have illustrated their advances in flexibility and electrical performance.^{26,27} Among various reported flexible batteries, metal–air based cells have attracted much attention for their high-power output and simple design.

To date, a wide variety of flexible energy storages based on metal/air batteries, including magnesium (Mg), aluminum (Al), and zinc (Zn), have been reported as power sources for flexible electronics.^{28–33} Among these metal/air batteries, Mg–air batteries have been considered as the most suitable candidates for wearable devices due to its good biocompatibility, excellent mass density, price competitiveness, and capability of exploiting aqueous electrolytes. To meet the integration requirements of these batteries with wearable electronics, the materials and devices of the batteries need to be bendable, stretchable, lightweight, and biocompatible.^{34–36} As a landmark type of soft and conductive material, hydrogels have manifested many advantages as electrolyte in numerous aspects to design wearable Mg–air batteries. First, in striking contrast to the conventional liquid aqueous electrolytes, hydrogel electrolyte is solid-state

and capable of eliminating the risk of liquid leakage under mechanical actions. Second, hydrogel-based Mg–air batteries can be easily imparted with excellent flexibility and mechanical strength, enabling great adaptability to the complex, dynamic environments. More importantly, hydrogels exhibit huge freedom in the design of polymer network for the regulation of ion mobility and transport, which has been proven as a powerful route to improve the electrochemical performances of Mg–air batteries. However, existing hydrogels-based Mg–air batteries normally suffer from inferior output performance as imposed by low ion conductivity of hydrogel electrolyte and unsatisfied anodic polarization, significantly restricting their applications in wearable electronic devices.^{37–39} In addition, although hydrogels are stretchable, the electrodes of Mg–air batteries are inherently rigid, leading to the poor stretchability of the whole hydrogels-based Mg–air battery device. Therefore, it is still challenging to exploit stretchable hydrogels-based Mg–air batteries with high power output performance. The question rises whether it is possible to combine the advanced mechanical designs of segmented layouts and deformable electrical interconnects with hydrogels-based batteries, which would be a new solution of power sources for wide applications in skin-integrated electronics and intelligent biomedical systems.

Here, we report a high-performance stretchable Mg–air battery based on biocompatible hydrogel as electrolyte. The battery can serve as a smart bandage for intelligent regulating of drug delivery in a self-powered and self-adaption manner for wound dressing, which is different from traditional treatment strategies. The good stretchability of the Mg–air battery relies on the stretchable dual-ions-conducting hydrogel (SDICH) that results from the meticulous choice of polymer skeleton and mobile ions, featuring high ionic conductivity as well as excellent stretchability and flexibility. To the best of our knowledge, this is the first reported stretchable Mg–air battery up to now. The stretchable Mg–air battery is capable of yielding a high energy capacity and power density of 9.9 mAh and 5.6 mW cm^{−2} with a high stretchability of 45%, which shows great output performance comparing to those published flexible batteries (Table S1 and Figure S1). Due to the advanced functional materials and well-designed mechanical structures, the battery could offer stable power output under various mechanical deformations, including stretching, twisting, bending, and crashing. By connecting two battery cells in series, the system could directly light up 120 light-emitting diodes (LED) concurrently under a wide range of temperature (from 8°C to 60°C) for more than 5 h. Further demonstrations of a self-adaption wound dressing system are shown in this work based on the integration of the stretchable battery with a passive feedback regulation sensor. The *in vitro* and *in vivo* results demonstrate that the drug delivery system can intelligently adjust antibiotic release rate and

dosage according to the real-time wound conditions, eliminating the use of complicated electronic monitoring devices, and thus allow to effectively prevent wound inflammation and promote wound healing. The fully stretchable Mg–air battery and self-adaption wound dressing show great potential in healthcare monitoring, medical treatment, and skin-integrated electronics.

2 | RESULTS AND DISCUSSIONS

Figure 1A presents the schematic illustration of the stretchable aqueous Mg–air battery based on the SDICH. The stretchable Mg–air battery consists of two layers of thin polydimethylsiloxane (PDMS, each thickness of 0.17 mm) at the top and the bottom, a graphene-coated nickel (G/Ni) foam (thickness of 1 mm), SDICH (0.5 mm), and thin Mg foil (0.05 mm). The soft and stretchable PDMS serves as the substrates and encapsulation layers simultaneously while the SDICH based on networks of polymer skeleton and mobile ions performs as the electrolyte and separator layer (Figure 1B). The polymer skeleton of SDICH synthesis associates with interpenetrating polymer networks of gelatin and poly(acrylic acid-acrylamide) (poly [AAc-AAm]), that offers great stretchability and flexibility. The mobile ions in the SDICH consist of choline chloride (ChCl) and sodium nitrate (NaNO₃), both of which play key roles in the performance improvement of Mg–air battery. Briefly, ChCl is capable of interacting with poly (AAc-AAm) by hydrogen bonding to provide high ionic conductivity for minimizing the internal resistance of the battery. On the other hand, NaNO₃ is free within polymer networks and beneficial for decreasing corrosion and anodic polarization in Mg and therefore enables high densities current output. As shown in Figures S2 and S3, the increase of NaNO₃ concentration in SDICHs results in obvious enhancement of conductivities, however, leads to a significant decrease in Young's modulus at the same time. This phenomenon is attributed to the more dissociation of original polymer chain interactions under high NaNO₃ concentration in the hydrogels.⁴⁰ Furthermore, high NaNO₃ concentration in SDICHs also causes the salting-out behaviors of polymers, that causes the collapse of polymer chains and forming of small pores,⁴¹ as evidenced by the macroscopical changes of hydrogels from transparency to opaque (Figure S2) and the microcosmic morphology changes from porous to dense structure (Figure S3).

On the account of the high Young's modulus, SDICH exhibits great stretchability and flexibility (Figures 1C and S4). Figure 1D,E shows the optical images of the cathode and anode of the stretchable battery, respectively. In the cathode, G/Ni foam performs as both the cathode catalyst and current collector (Figure 1D). The

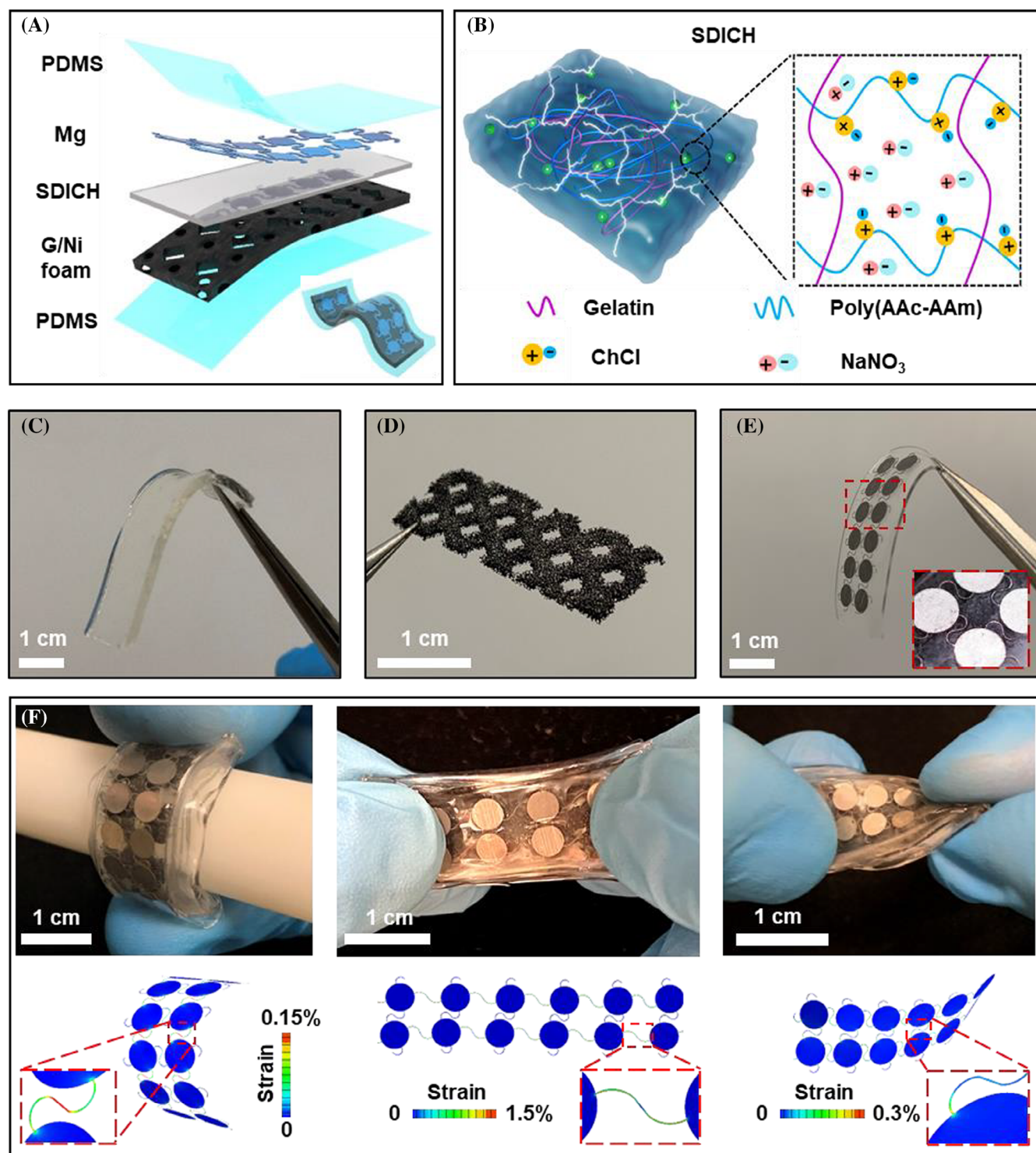


FIGURE 1 Stretchable Mg-air battery based on the SDICH. (A) Schematic illustration of a fully stretchable Mg-air battery. (B) Schematic illustration of the SDICH, with material bonding structures of the hydrogels. (C-E) Optical images of the electrolyte, cathode and anode layer of the battery. Enlarged image in (E) details the mechanical design of the anode. (F) Optical images of the stretchable battery under bending, stretching and twisting and their corresponding finite element analysis (FEA).

ballistic transportation properties of graphene offer the electrode good conductivity and thus can act as a promising catalyst for oxygen reduction reaction, while the nickel foam with porous structure and remarkable

electrical conductivity allows gas diffusion and current collection simultaneously. As a result, the surface resistance of the cathode can be as low as 38.6 Ω (Figure S5). In addition, diamond-hollow structure design in G/Ni

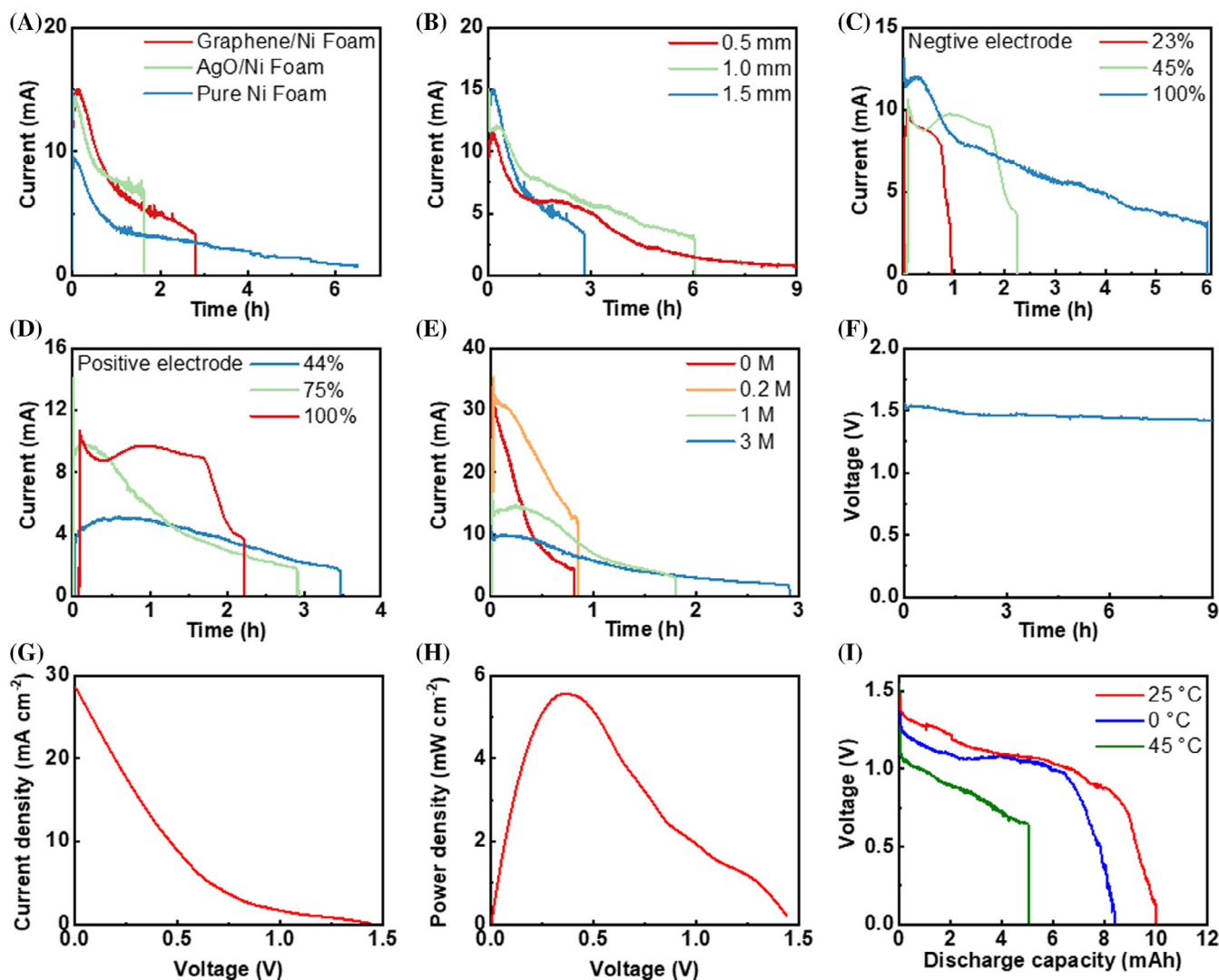


FIGURE 2 Electrical characterizations of the stretchable Mg-air battery. (A–E) Short-circuit performance of the stretchable battery based on various cathode materials (A), electrolyte layer thickness (B), anode area ratios (C), cathode area ratios (D) and NaNO₃ concentrations (E). (F) Open circuit output of the stretchable battery as a function of time. (G) and (H) The polarization curve and power density curve of the stretchable battery. (I) Discharging capacity of the stretchable battery at different operation temperatures, where the discharging current density is 1 mA cm⁻².

foam is able to further enhance the stretchability of the cathode. Besides, as shown in Figure S6a, the oxygen reduction reaction (ORR) current of G/Ni foam is much higher than that of Ni foam in O₂-saturated atmosphere, which is caused by the larger specific surface area and better conductivity of graphene.^{42,43} The G/Ni foam electrode also exhibits superior ORR catalytic activity to Ni foam electrode in terms of more positive onset potential and half-wave potential ($E_{1/2}$).^{44,45} The G/Ni foam shows typical LSV curves of ORR in O₂-saturated atmosphere, which cannot be found in the N₂-saturated atmosphere, demonstrating the ORR activities of the electrode (Figure S6b) is great and in well consistent with the reported results.^{46–48} Figure 1E shows the Mg anode with

island-bridge interconnecting geometries on a stretchable PDMS substrate. The mechanics of the serpentine interconnection and round shape islands enable improved stretchability of the Mg electrodes. Therefore, the stretchable nature of all the components allows the battery to exhibit robust mechanical properties, which could be bent, stretched, and twisted to accommodate natural body joint motions (Figure 1F). In addition, the battery could be directly worn on different human body parts (like cubital fossa), as shown in Figure S7, demonstrating its stability and comfortability for users during various practical applications. Finite element analysis (FEA) guided the design of the serpentine interconnection, which is the first part to reach the yield strain (the yield

strain of Mg, 1.5%). It can be seen that the strain of the Mg foil is much lower than the yield strain when the device has bent and twisted deformation under the working state without damage. The device can be stretched to 45% under the yield strain of Mg. The stretchability can be further enhanced by decreasing the feature size of the serpentine electrode structure or manufacturing electrode into kirigami structure.⁴⁹ Additionally, the lower electrode covering area ratio can also help to achieve greater stretchability for the battery, as shown in Figure S8.

Figure 2 shows the systemically studies of the electrical characteristics in the stretchable battery from five aspects, including the catalyst type of oxygen reduction (Figure 2A), the thickness of the SDICH/3 M NaNO₃ (Figure 2B), electrode areas ratios (Figure 2C,D), and the NaNO₃ concentration in SDICH (Figure 2E). As shown in Figure 2A, the battery (SDICH/3 M NaNO₃, 1.5 mm) adopting the G/Ni foam can continuously yield a high average short-circuit (SC) current of 9.2 mA for 2.8 h, which are no doubt benefited from the excellent catalytic effects on oxygen reduction reaction in graphene, and effective electron transfer behaviors between graphene and the current collector (Ni foam). Figure 2B shows the influence of thickness of SDICH layer on the performance of the battery. Considering both SC current and operation time, it can be seen the battery with 1 mm SDICH/3 M NaNO₃ owns the best electrical performance with the average SC current and operating time of 7.6 mA@6 h, while the performance of the batteries with 0.5 and 1.5 mm SDICH are 6.08 mA@9 h and 9.2 mA@2.8 h for 1.5 mm SDICH, respectively. The results are led by thinner electrolyte layer typically increase the risk of internal short current for the battery,^{50,51} while thicker SDICH can accelerate Mg electrode corrosion rate due to the increasing internal resistance and amounts of nitrate ion. As a result, it can be seen that 1 mm is the optimized thickness of the SDICH/3 M NaNO₃ electrolyte layer for this type of battery. To achieve great stretchability, two electrodes of the battery need to be patterned into stretchable formats (serpentine designs for the Mg layer and diamond-hollow structure for the G/Ni foam). Figure 2C,D shows the electrical responses of the battery with 1 mm thickness of SDICH/3 M NaNO₃ and graphene/Ni foam as a function of area ratio of the two electrodes. It is obvious that the larger electrode area ratio could contribute greater SC current but reduce the stretchability of the battery. To achieve high stretchability, the negative and positive electrodes adopt serpentine designs and hollow structures, respectively (Figure 1F,G). Lower electrode area ratio could realize high stretchability, which is proven by the mechanical stimulation in Figure S8. Considering the power output and operation time, we adopt the negative electrode (Figure S9) and

positive electrode (Figure S10) with 45% and 75% area ratio, respectively, which could ensure both sufficient stretchability for most practical applications and good output performance. As a result, the optimized battery could yield an average SC of 5.8 mA with an operation time over 2.9 h. Figure 2E shows the variations of the electrical performance of the battery on SDICHs with different NaNO₃ concentrations. It is found that the battery with SDICH/0.2 M NaNO₃ could generate the highest average SC of 22.2 mA for 0.8 h, compared with the 19.2 mA@0.8 h for 0 M NaNO₃, 8.8 mA@1.8 h for 1 M NaNO₃, and 5.7 mA@2.9 h for 3 M NaNO₃, which is consistent with the reports of low concentration of electrolytes contributing greater electrical performance of Mg-air batteries.³⁹ Figure 2F presents the open-circuit (OC) voltage of the battery with optimized parameters, that associates with stable output of 1.44 V for 9 h. The polarization curve of the battery in Figure 2G shows that the maximum current density could reach 28.6 mA cm⁻². Accordingly, the maximum power density is calculated as 5.6 mW cm⁻², which is a considerable value as power output in the stretchable Mg-air batteries (Figure 2H). Following, the discharging capacity of the optimal battery as a function of operating temperature is measured. As shown in Figure 2I, the battery shows a high capacity of 9.9 mAh at 25°C, while the capacity would decrease to 4.9 mAh when temperature raise to 45°C, which is caused by the faster dehydration of the SDICH with the increasing operating temperature. And the low temperature has slight negative effects on the battery capacity with 8.8 mAh at 0°C (Figure 2I). By sealing the battery away from air, the battery could be stored for 7 days.

To demonstrate the stability of the stretchable Mg-air battery, the electrical responses of the battery under various mechanical deformations, including twisting, bending, stretching, and crashing had been investigated (Figure 3). Figure 3A-C shows that the OC voltage outputs of the Mg-air battery under twisting (0°–90°), bending (0°–90°), and stretching (0%–20%) do not reveal obvious fluctuations, illustrating the battery has excellent flexibility and stable outputs under various operation status. The decrease of the voltage of the battery stretched to 40% is caused by the displacement of adjacent layers induced by the large Young's modulus differences among each layers inside the battery. While the voltage can recover to its initial value after releasing the stretch for the battery. Furthermore, a continuous stretching testing is conducted to prove the stability of the battery, as shown in Figure 3D. By continuously stretching the battery to 30% at a frequency of 3 Hz, the OC voltage of the battery can be stabilized to around 1.44 V (Figure 3E,F) due to the excellent mechanical designs and stretchability. Here, the remarkable stretchability and stability of

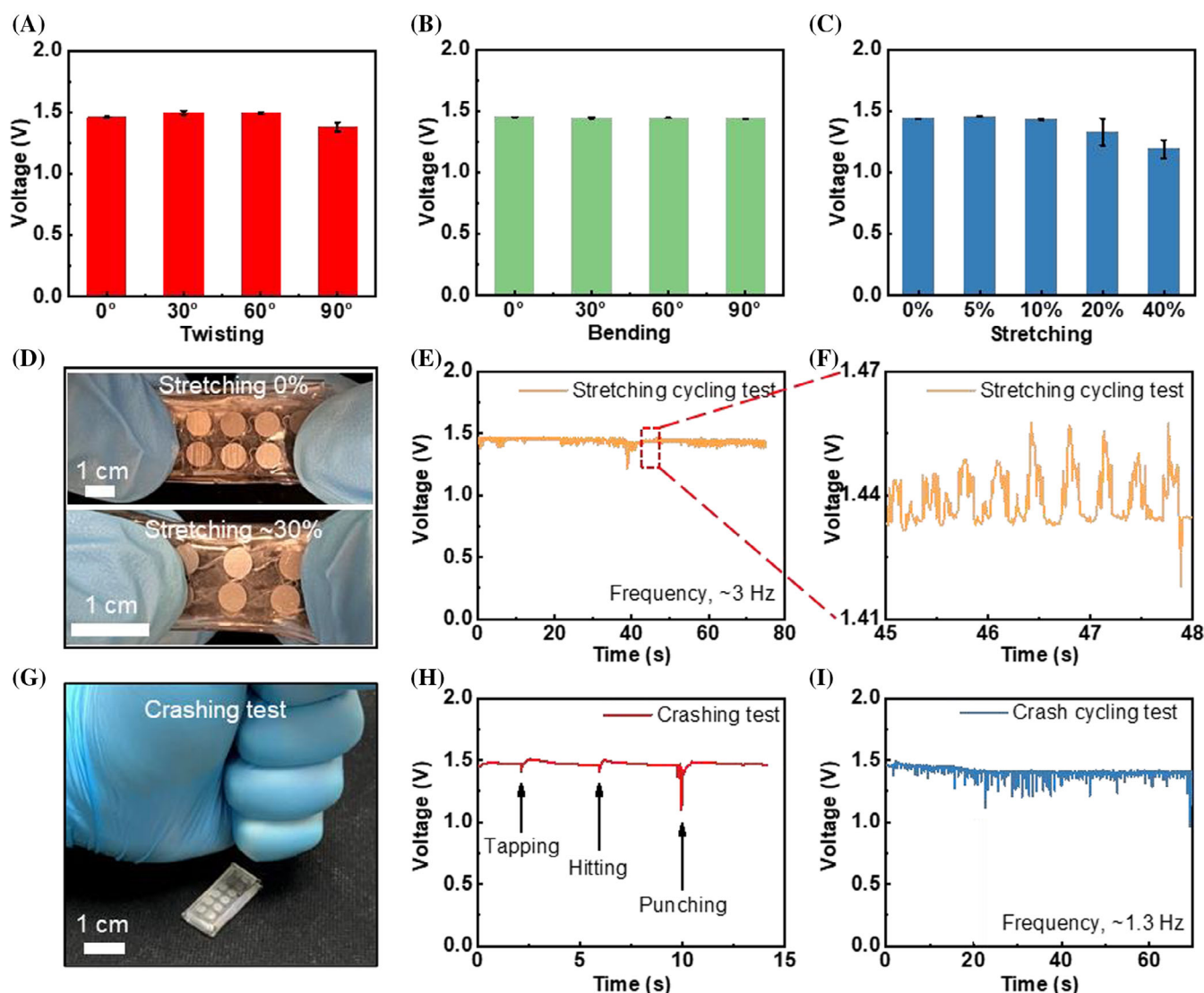


FIGURE 3 Mechanical characterization of the stretchable Mg-air battery. (A) to (C) Open circuit output of the stretchable battery under twisting from 0° to 90° (A), bending from 0° to 90° (B), and stretching from 0% to 40% (C). (D) Optical images of the stretchable battery under stretching of 0% and 30%. (E,F) Open circuit output of the stretchable battery under bending with a frequency of 3 Hz and the partial enlarged details. (G) Optical image of the stretchable battery under hard crashing test. (H,I) Open circuit output of the stretchable battery crashed with a frequency of 1.3 Hz and the partially enlarged details.

the SDICH with 0.2 M NaNO_3 are shown in Figures S11 and S12. The SDICH/0.2 M NaNO_3 can be stretched up to 1000% while there is negligible conductivity attenuation after 100 continuous stretching cycles at 200%. Figure 3G presents a crashing test associating with hitting the battery repeatedly under different applied forces. Here, the impact forces are 10 N@ tapping, 20 N@ hitting, and 50 N@ punching. By increasing the hitting force, although the momentary fluctuation of the OC voltage drops from 1.41 to 1.13 V, then it could recover to 1.44 V (Figure 3H). Figure 3I shows the electrical response of the battery during the 70 s continuous tapping test at a constant frequency of ~ 1.3 Hz, the results

indicate that the battery could yield stable outputs even in harsh environments. Here, the elastic deformation of the G/Ni foam and low Young's modulus of the SDICH enable the fast recovery of the battery under various mechanical deformations.

To prove the potential for practical applications under mechanical deformations (Figures S13 and S14), two battery cells are connected in series yielding high voltage output that can directly power 120 green LEDs under various mechanical deformations, including bending 90° (Figure 4A), twisting 60° (Figure 4B), and stretching 20% (Figure 4C). As a result, the two battery cells could keep enlightening these LEDs array under

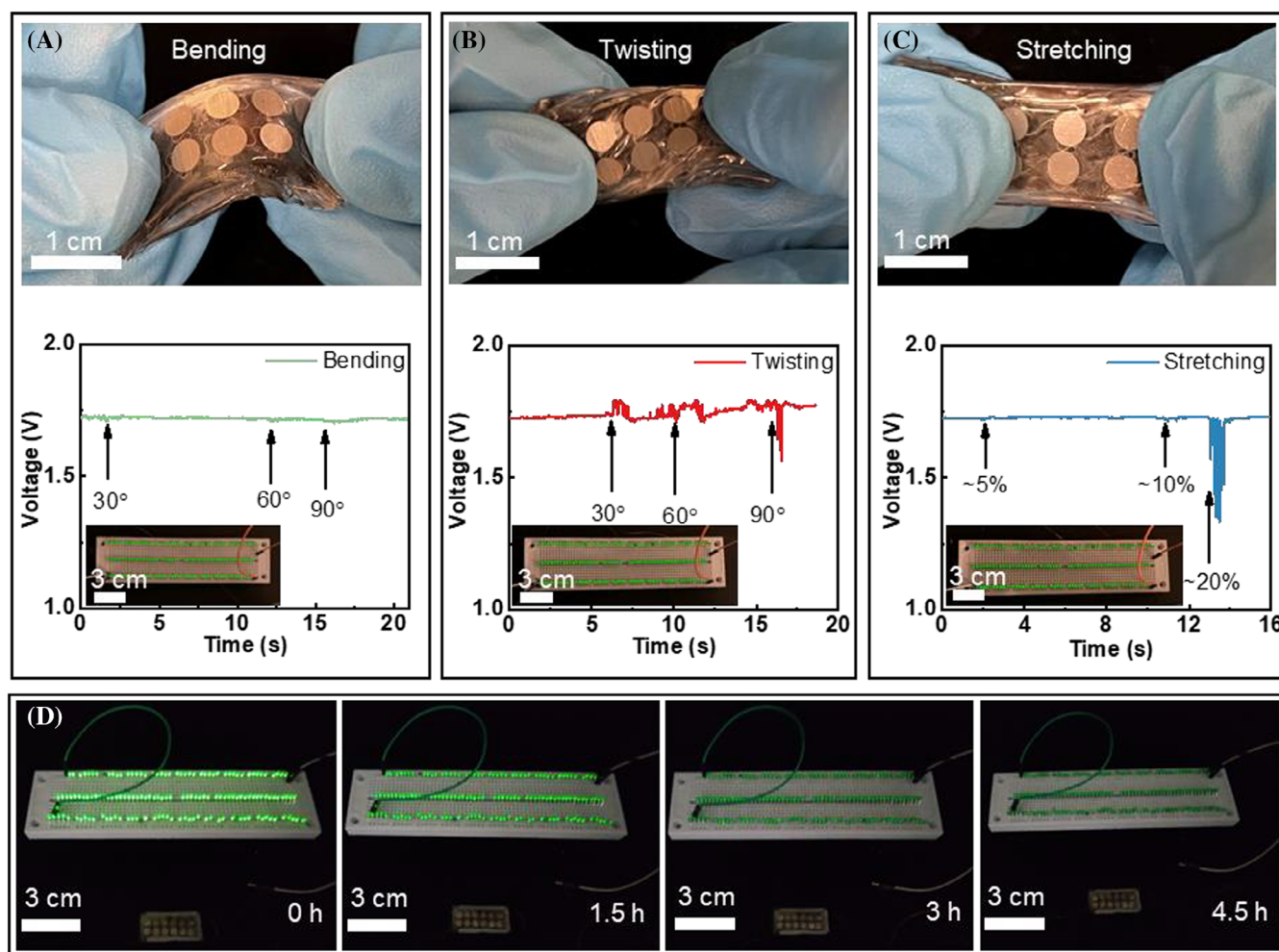


FIGURE 4 Load characterization of the stretchable Mg-air battery. (A–C) Optical images of two stretchable batteries connected in serials to light 120 green LEDs under bending (A), twisting (B) and stretching (C), and the related voltage outputs plots of these LEDs. (D) Optical images of the lightness changes of 120 green LEDs powered by the stretchable batteries within 4.5 h under room temperature.

various deformations with a stable voltage of ~ 1.73 V (Figure 4A–C and Movie S1). Here, the voltage fluctuation may result from the relative displacement of adjacent layers which is induced by the large Young's modulus difference among each layer of the battery. As shown in Figure 4D, the battery cells connected in series could support the LEDs lighting for up to 5 h due to their excellent electrical characteristics. As shown in Figures S15 and S16, the voltage and current output of the batteries are continuously dropping along with operation time, inducing the decreasing lighting intensity of the 120 LEDs. In addition, the battery cells could operate and power these LEDs under various ambient temperatures ranging from 8°C to 60°C with stable voltage and lightness (Figure S15). Under the temperature of 8°C and 21°C , the operation times for 120 LEDs can be as long as 5 h, while those of 40°C and 60°C would decrease a lot led by the dehydration of the SDICH.

Moreover, we combine the biocompatible batteries and flexible drug delivery system for self-adaption wound dressing. Here, we prepare a drug-loaded hydrogel by mixing antibiotics into the hydrogel serving as the drug container for wound healing. And a thermistor works as the sensor for the self-adaption system (Figure 5A). Figures 5B, S17, and S18 present the components of the whole drug delivery system, containing two battery cells working as the power source, two layers of stretchable copper traces-based electrodes (Figure S17) for applying external voltage onto the drug-loaded hydrogel, a layer of drug-loaded hydrogel, and bottom encapsulation layer with a thermistor embedded. Based on the iontophoresis induced by the battery,⁵² antibiotics in the drug-loaded hydrogel could be transported to the target injury area along with the flow of current by attaching the device directly to the epidermis (Figure 5A). Furthermore, when temperature rise caused by the inflammation of wounds is detected, a high voltage state in the battery would be

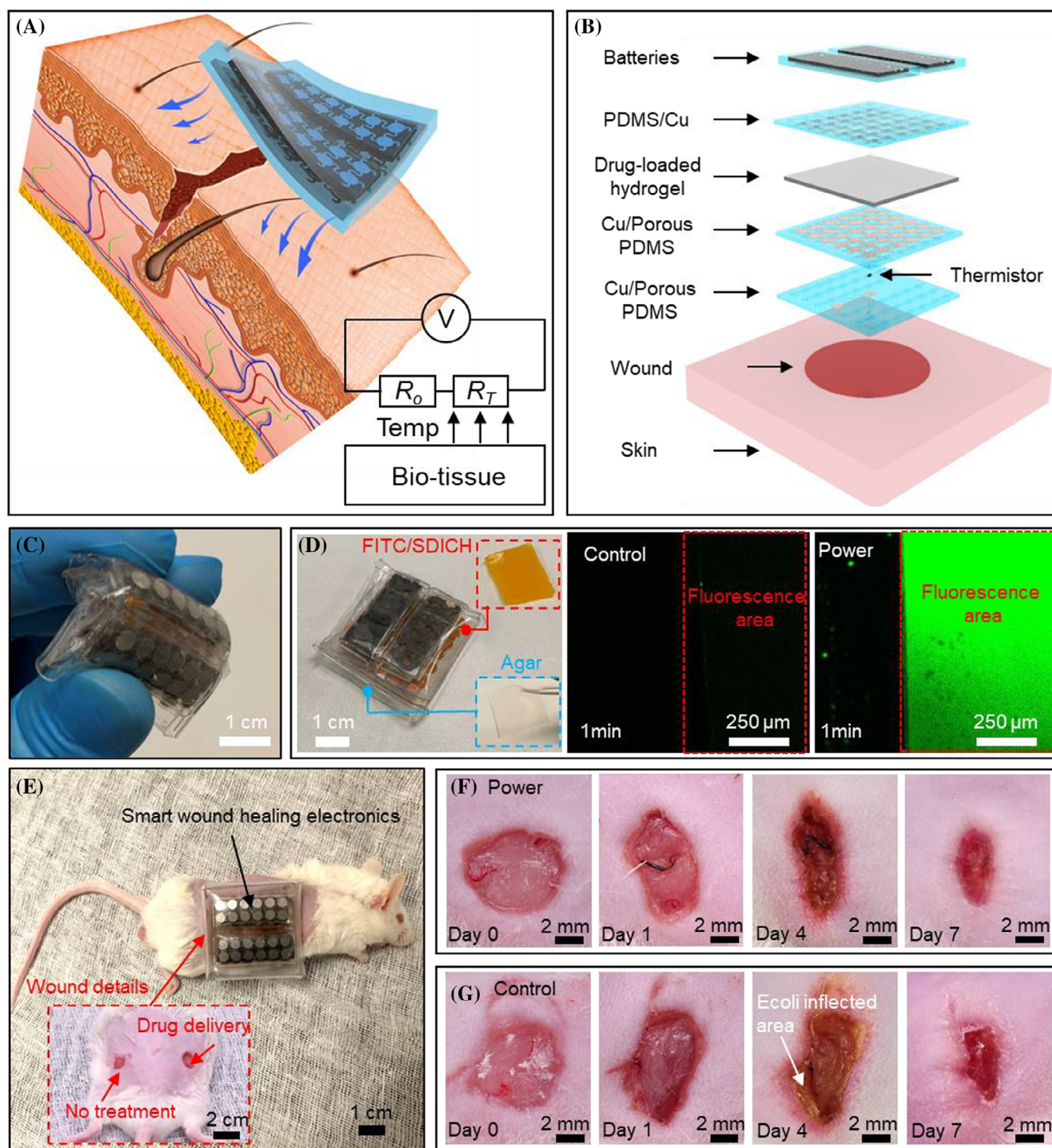


FIGURE 5 Stretchable self-adaption wound dressing system. (A) Concept and working principle of the self-adaption wound dressing system. (B) Schematic illustration of the self-adaption wound dressing system. (C) Optical image of the self-adaption wound dressing system. (D) Optical images of the in vitro test and corresponding fluorescence images in agars of control and experiment groups. (E) Optical image of the in vivo test for rat model. (F,G) Optical images of the wound changes in the control and experiment groups.

accomplished due to decreasing resistance of the thermistor (Figure S19). As a result, more antibiotics can be delivered to the target area and escalate the healing process (Figure S20). As every component is flexible and stretchable, the whole system exhibits excellent flexibility

and wearability (Figure 5C). To characterize the self-adaption wound dressing system, in vitro tests for agar and in vivo tests for mice models have been conducted. As shown in Figure 5D, fluorescein isothiocyanate (FITC), that is, a small molecule substance is exploited to

replace antibiotics for examining the electrophoresis effects. Compared with pure agar (Figure S21), few FITC molecules could permeate into agar within a minute through molecular diffusion. When power sources are connected on to the drug-loaded hydrogel, it could be observed that FITC molecules could quickly pass into agar by electrophoresis effects and exhibits a much larger fluorescence area and fluorescence intensity, demonstrating the validity of the system for promoting the release of small molecule drugs. Furthermore, we select the rat model for in vivo test (Figure 5E). For obtaining noticeable contrasting results, 20 μl *Escherichia coli* (10^8 CFU ml^{-1}) is smeared onto each wound surface, respectively. As shown in Figure S22, the voltage on the drug-loaded hydrogel would increase with the temperature rising. As a result, more antibiotic would be released by a faster rate. Compared with the wound wearing with the self-adaption system (Figure 5F), there is an apparent *E. coli* propagation phenomenon around the no-treated wound after 4 days (Figure 5G). By analyzing the wound healing area size, it is apparent that the wound healing rate of treated sample is faster than that of the no-treated wound, as shown in Figure S23. Furthermore, although inflammatory responses are inevitable due to the smeared large amounts of *E. coli*, the amount and density of inflammatory cells in infiltration area of control sample are both much higher than those of treated sample, demonstrating excellent wound healing effects of the self-adaption wound dressing system (Figure S24). As a result, the in vivo test results prove that the self-adaption wound dressing system has excellent therapeutic effects on wound healing and avoiding wound infection,⁵³ which provides a feasible method for intelligent wound dressing and personalized medicine (Table S2).

3 | CONCLUSION

In summary, we have developed a fully stretchable Mg-air battery based on the SDICH, which is constructed by a cross-linked network of gelatin and poly (AAc-AAm) and the mobile ions of ChCl and NaNO_3 with great stretchability and conductivity. The optimized battery yields a high OC voltage of 1.44 V, an energy capacity of 9.9 mAh, and a maximum power density of 5.6 mW cm^{-2} . Following, two cells connected in serials are able to light 120 green LEDs over 5 h. Advanced materials and mechanics designs allow the assembled Mg-air battery to exhibit stable electrical outputs under various types of deformations, for instance, maintaining unchanged electrical performance under 45% stretching. Furthermore, a soft and wearable self-adaption wound dressing system was constructed based on the biocompatible and

stretchable battery. The in vitro and in vivo tests both demonstrate the system can effectively and intelligently deliver drugs to the wound and adjust the drug amounts spontaneously according to the wound healing condition. The whole system is compact with excellent stretchability without any additional power supplies. The results exhibit very promising prospects for advancing stretchable electronics, for instance, intelligent wound dressing and personalized health monitoring.

4 | EXPERIMENTAL SECTION

4.1 | Synthesis of the SDICH

SDICH was synthesized by a facile photo-triggered gelation procedure, similar to our previous reports.^{54,55} Briefly, gelatin (10 wt%) was first added into 10 ml deionized water and stirred at 60°C for 2 h. Then, 1 g acrylic acid (AAc), 1 g ChCl, 3 g acrylamide (AAm), 20 mg *N,N'*-methylenebisacrylamide (MBAA), 90 mg 2-hydroxy-4'-(2-hydroxyethoxy)-2-methylpropiophenone (photoinitiator 2959), and NaNO_3 (0, 0.2, 1, and 3 M) added into the above gelatin solutions. After being degassed for 30 min, the mixed solution was placed at illumination of UV light for 10 min to generate hydrogels.

4.2 | Fabrication of the stretchable Mg-air battery

The electrode fabrication commenced with laser cutting Mg foil (thickness, 50 μm) and Ni foam (thickness, 1 mm) into batteries electrodes. Then, the Mg electrode was picked up by the water-soluble tape, followed by deposition of adhesive layers (Ti/ SiO_2 , electron-beam evaporation, 5/50 nm thick). The patterned Ni foam was immersed into graphene ink and then baked on a 100°C hotplate for 5 min to fabricate the G/Ni foam. The encapsulation of the Mg-air battery was conducted on a glass substrate, which was firstly cleaned by acetone, ethanol, and deionized water (DI water, > 18 $\text{M}\Omega\text{ cm}$) sequentially. Sodium stearate aqueous solution was spin coated on the glass sheet, serving as a thin sacrificial layer, then dried under 100°C for 5 min for later releasing steps. Following, PDMS (30:1) was spin-coated on the sacrificial layer at 600 rpm min^{-1} for 30 s, cured on a 100°C hotplate for 20 min. After that, the Mg electrode on the water-soluble tape was transfer-printed on the PDMS substrate, and the SDICH layer and the G/Ni foam were put on the Mg electrode lay-by-layer. Finally, the battery was packaged by the bottom PDMS film with preserved

holes on its four corners, as shown in Figure 3G. The overall thickness of the battery is 2.4 mm.

4.3 | Development of the self-adaption wound dressing system

The fabrication process of the drug-loaded hydrogel was similar with that of SDICH. The only difference is that the NaNO_3 is replaced by penicillin (1 mg ml^{-1}) or FITC (1 mg ml^{-1}). The copper pattern on the PDMS was also obtained by laser cutting. First, the substrate was obtained by spin-coating PDMS (30:1) on a glass at 600 rpm min^{-1} for 30 s. And then a Cu/PI film ($18 \mu\text{m}$) was stuck on the PDMS before baking on a 100°C hotplate for 20 min. Next, the Cu/PI film was patterned by laser cutting. After that, using hole puncher (2 mm diameter) to get the porous PDMS. At last, the system was capsulated by the bottom PDMS after putting every component layer-by-layer.

4.4 | Characterization

Electrical resistance of SDICH was measured by Keithley 2400 sourcemeter. Mechanical performance of SDICH was investigated using a servo control computer tensile testing machine (TH-8203 S, China) with a 100 N load cell. The open-circuit voltages of batteries were measured by PowerLab (PL3516/P 16/35) with a constant sampling frequency of 400 Hz. The current was obtained by monitoring voltages of the 10Ω resistance connecting to the battery. The polarization curve was measured by linear sweep voltammetry from open-circuit voltage to 0 V by electrochemical workstation of CHI 660E. The current density was calculated by dividing the current by electrode area, and the power density was obtained by current density multiple voltage. The fluorescence images were obtained by the confocal laser scanning microscope of LEICA TCS SP8.

4.5 | Mechanical analysis

The FEA commercial software ABAQUS (Analysis User's Manual 2020) was used to calculate the bent, stretched, and twisted of the structure. The structure was modeled by 2.86 million hexahedron elements (C3D8R). The minimal element size was 0.001 mm, which ensured the convergence and the accuracy of the simulation results. The maximum yield strain of Mg is 1.5%, which is the first part to reach the maximum yield strain. The elastic modulus (E) and Poisson's ratio (ν) used in the analysis

were $E_{\text{Mg}} = 45 \text{ GPa}$, $\nu_{\text{Mg}} = 0.28$, $E_{\text{PDMS}} = 67 \text{ kPa}$, $\nu_{\text{PDMS}} = 0.49$, $E_{\text{Hydrogel}} = 23.3 \text{ kPa}$, $\nu_{\text{Hydrogel}} = 0.49$, $E_{\text{G/Ni foam}} = 1.4 \text{ MPa}$, $\nu_{\text{G/Ni foam}} = 0.3$.

4.6 | In vivo experiments

The ICR mice was firstly anesthetized by 0.2 ml chloral hydrate aqueous solutions (10%) before operation. And then using skin punch (diameter, 6 mm) to obtain two wounds on the back area of the mice. Following, smearing $20 \mu\text{l}$ *E. coli* (10^8 CFU ml^{-1}) on every wound surface. For the experiment group, attaching the self-adaption flexible drug delivery system for treatment. Frozen tissue sections were subjected to HE staining for proving the efficiency of the effects of the self-adaption flexible drug delivery system. Animal experiment was in accordance with the institutional guidelines and approved by the Research Committee of City University of Hong Kong.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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